Test 4 Study Plan

Thursday, November 12 · last 30 minutes of class · 20 marks (+bonus) · closed book, no calculator

1. What Test 4 covers

The README pairs **PN4 (Decision Making, Learning, and the LMU)** with Test 4, and PN4 says the test "will assume you can do everything above by hand or from scratch." Each notebook section maps onto course content:

PN4 section	Lecture source	What you must be able to do on paper
§1 Accumulate-to-threshold decision	L06 integrator ($A' = \tau A + I$); radius and intercepts (L03); function decoding (L05)	Derive the implemented ODE; sketch ramps and saturation; explain saturation through decoded feedback beyond the radius; noise as drift-diffusion; decode a step and its limits; the intercept trick; 1-D vs 2-D ensembles
§2 Legendre Memory Unit	L07 (all) + LMU overview slides	Sampling/Nyquist; basis functions and Legendre coefficients; derive small LMUs ($q = 1$ is a low-pass filter); A', B'; decode delays with $\tilde{P}_i(\theta'/\theta)$; why a window classifies frequency; q and neuron sweeps
§3 PES learning	L08 supervised half	Derive the delta rule and PES; error sign; network diagram; learning-rate and duration trade-offs; why error plateaus above 0
§4 Adaptive control	L08 adaptive-controller slide	PD steady-state offset; what is learned ($mgL\cos\theta$); why the PD output is the error, and its sign; sketch PD vs adaptive
§5 SVD analysis	L09 (all)	Which matrix (Γ) and why; spectrum and noise floor; χ basis $\approx$ Legendre; noise derivation; $S_m > \sigma^2$; multiset count
Lower priority	L08 unsupervised half	Hebb (unstable), Oja ($\rightarrow$ first PC), BCM, STDP window, PCA = covariance eigenvectors. Covered in lecture but not in PN4: know the definitions and sketches.
Not on Test 4	L10 Symbols	That is PN5 and Test 5.

2. Schedule

Tonight (Wed), about 3.5 h with breaks

1. **20 min, orient.** Skim the PN4 *Expected* checkpoints and discussion prompts. Read L07 "Implementing Delays," L08 "Delta" and "PES" (plus the adaptive-controller slide), and L09 "Noise."
2. **60 min.** Guided Assignment, Stages 1 (decision) and 3 (LMU).
3. *Break, 10 min.*
4. **75 min.** Stages 4 (PES), 5 (adaptive control), 7 (SVD).
5. **20 min.** Stages 2 and 6: attempt quickly, then read the solutions.
6. **10 min.** On a blank page, write the recall sheet below from memory. Check it, then sleep.

Tomorrow morning, about 75 min

1. **30 min.** Practice Test 4, timed and closed book, with no calculator. Do the bonus only if time remains.
2. **20 min.** Mark it with the key. For each lost half-mark, write the missing *mechanism*, *assumption* or *labelled feature*.
3. **15 min.** Stage 8 readiness check for any topic still shaky.
4. **10 min, right before class.** Redraw the six key sketches from memory.

If short on time, skip Stages 2 and 6 and the bonus. Do not skip Stage 7: the SVD was taught on Nov 10, so it is the freshest and least practised material, and it may be on the test.

3. Recall sheet (be able to write all of this cold)

- Matched synapses: $A' = \tau A + I$, $B' = \tau B$. An input with transform 1 into an identity-recurrent ensemble integrates at $1/\tau$, so $x = ut/\tau$.
- Saturation: decoders are fit only on $[-1, 1]$, so $\hat{f}(x) < x$ beyond it; the equilibrium satisfies $x - \hat{f}(x) = B'u$.
- The intercept dead zone makes the output *exactly* 0 (all neurons silent, linear decode).
- Nyquist: $N = 2BT$. Orthonormal coefficient: $x_i = \langle \phi_i, f \rangle$.
- Legendre: $\tilde{P}_0 = 1$, $\tilde{P}_1 = 2r - 1$, $\tilde{P}_2 = 6r^2 - 6r + 1$. Values: $\tilde{P}_i(0) = (-1)^i$, $\tilde{P}_i(1) = 1$.
- LMU: $u(t - \theta') \approx \sum_i \tilde{P}_i(\theta'/\theta) x_i$. For $q = 1$, $\dot{x} = (u - x)/\theta$.
- Delta rule: $\Delta d_j = -\eta \varepsilon a_j$ with $\varepsilon = \hat{y} - y^d$.

- PES: $\Delta w_{ij} = -\eta \alpha_i \langle \mathbf{e}_i, \boldsymbol{\varepsilon} \rangle a_j$.
- PD offset: $\theta_{ss} \approx \theta^* - mgL\cos\theta^*/k_p$. Adaptive torque $\to mgL\cos\theta$; error $= -\tau_{PD}$.
- Hebb $\Delta w = \eta a_i a_j$. Oja $\eta(a_i a_j - a_i^2 w_{ij})$. BCM $\eta a_j a_i(\theta - a_i)$.
- $D = \Gamma^{-1}\Upsilon$ with $\Gamma = A^T A$ (neurons × neurons), $\Gamma = USU^T$, $\chi = AU$, $\Phi = U^T D$.
- $\Phi_n = \langle \chi_n x \rangle / (S_n + \sigma^2)$. Usable modes: $S_m > \sigma^2$.
- Count of order-l terms in D dimensions: $N_S = \binom{l+D-1}{l}$.
- PES modes relax at rate ηS_m, so high-order functions learn slowly.

4. Sketches to rehearse (label axes, thresholds, values)

(1) Accumulator ramps for $\pm 0.1, \pm 0.2$ that saturate just beyond ± 1, and $\hat{f}(x)$ vs x. (2) A noisy drift-diffusion trace that hits a bound. (3) choice(x) ideal vs decoded, plus tuning curves with intercepts in $[0.4, 0.9]$. (4) LMU: step response of $q = 1$ vs a true delay; RMSE vs q (U-shape) and vs neurons (decreasing). (5) PES chunk-RMSE curves for two learning rates; pendulum PD vs adaptive. (6) Singular spectrum on a log axis with the σ^2 floor; χ_1–$\chi_4 \approx$ Legendre; STDP window.

5. How past tests were marked, and how to answer

Tests 1–3 rewarded *mechanism + assumption + observable consequence*, labelled sketches and short symbolic derivations far more than arithmetic. For every "explain" item, name the cause (for example, decoders fitted only on the radius), the condition (matched synapses, fixed neurons) and what you would see. Budget about 1.5 minutes per mark. Do sketches immediately, because they are cheap marks, and leave the bonus for last.

6. Source traps found while preparing

- **PES sign:** the L08 derivation uses $\varepsilon = \hat{y} - y^d$ with $\Delta d = -\eta \varepsilon a$, while one figure caption writes $\varepsilon = y^d - y$. Either works if the update sign matches. Say which convention you use.
- **PN4 §1:** the input is *not* scaled by τ, so the accumulator integrates 10× faster than $\dot{x} = u$. Saturation is near ± 1 because of the radius, not because of a designed threshold.
- **L09** writes A as samples × neurons, so $\Gamma = A^T A$. PN4 uses neurons × samples, so $\Gamma = AA^T$. Either way Γ is $N \times N$.
- **L08 STDP:** the depression branch should be $-\alpha^- e^{+\Delta t/\tau^-}$ for $\Delta t < 0$; the notes' exponent sign is a typo. The BCM sign in the lecture is the homeostatic version.